

Hydrodynamic and Radiative Modelling of Colliding Wind Nebulae

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Context

There are a handful of Wolf-Rayet (WR) colliding wind binaries (CWBs) in the Galaxy – binary stellar systems where stellar winds collide to copiously form dust in the shocked, turbulent material. Systems like these formed **the first carbon dust in the Universe**, and modelling them helps us understand how they have shaped the Galaxy.

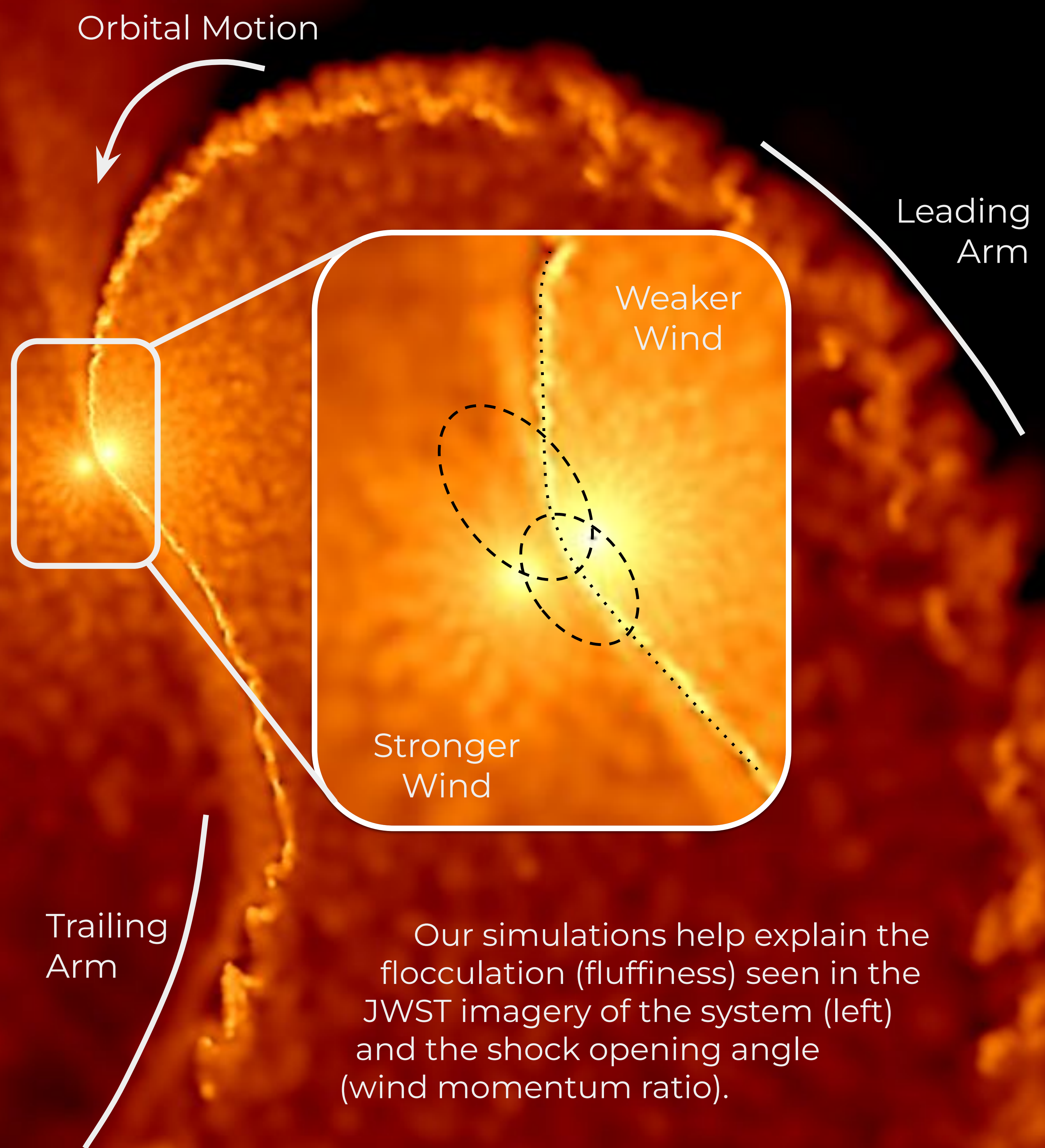
Hydrodynamic Modelling

The Apep system (pictured below) is the **most unique WR CWB**: it is the only dusty CWB to have two WR stars¹, its orbital period is an order of magnitude longer than the next longest CWB², and it is the only **confirmed trinary** whose O supergiant carves a cavity in the dusty nebula³.

We use the smoothed particle hydrodynamics code PHANTOM to model the colliding wind shock and **dust nucleation in Apep for the first time**. We use results from geometric models of the nebula³ to reproduce the observed nebula phenomenology: the simulation on the right of the poster shows the gas density cross-section of the system, where **wind collision results in two shocked arms** that are **wrapped into a spiral** by the orbital motion.



Han et al., White et al., Alyssa Pagan, NASA/ESA



Our simulations help explain the flocculation (fluffiness) seen in the JWST imagery of the system (left) and the shock opening angle (wind momentum ratio).

We have validated the inner binary with our hydrodynamic simulations, and we now aim to include the tertiary to see how it interacts with the dust plume as proposed in our published paper.

Radiative Transfer



We are able to recreate the JWST image of Apep by simulating radiative transfer through the dust geometry of our model. This ray-traces photons and captures the emission and scattering processes that happen in reality.

Using **MCFOST together with our geometric models** allows us to **simulate Apep and other CWBs in a fraction of the time** it would take a hydrodynamics code. Doing this allows us to

- estimate the **dust mass** of the nebulae,
- infer the dust **grain size distribution**, and
- predict **infrared light curves** of CWBs.

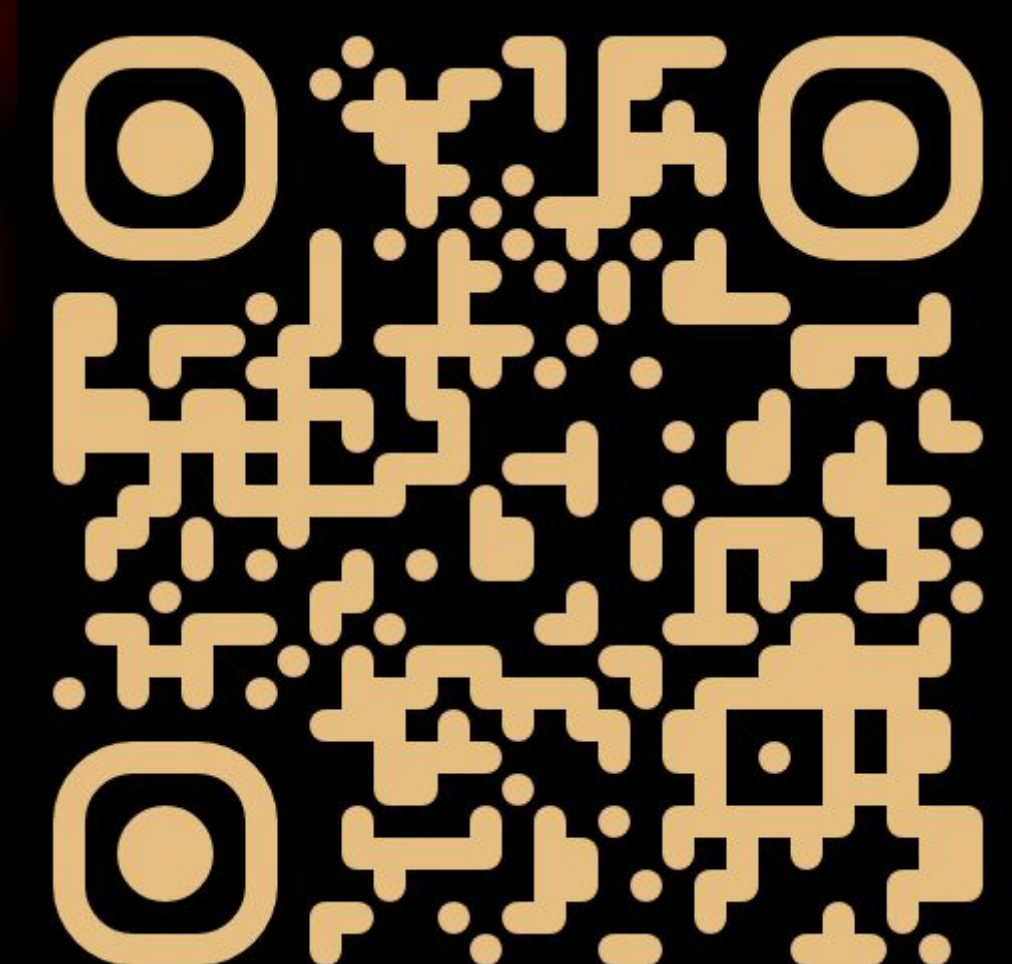
By using what we learn on already-imaged systems, we may eventually be able to model far away systems that can never be directly imaged.

Summary

We are modelling the Apep system with hydrodynamic and radiative transfer codes to understand dust formation and destruction in one of the most puzzling triple systems known – a modern analogue to early dust formation in the Universe.

Our results are in good agreement with observations, and further work

- 1) including the tertiary in the hydrodynamic simulations, and
- 2) interfacing geometric models with radiative transfer codes will allow us to **shed light on the processing of dust in the Galaxy**.



Scan to see our Apep paper!

Gas Density (g cm^{-3})
 10^{-15} 10^{-12} 10^{-9}

1. Callingham et al (2020) 'Two Wolf-Rayet stars at the heart of colliding-wind binary Apep' MNRAS, 495, 3323
2. Han, White et al (2025) 'The formation and evolution of dust in the colliding-wind binary Apep revealed by JWST' arXiv:2507.14498
3. White et al (2025) 'The Serpent Eating Its Own Tail: Dust Destruction in the Apep Colliding-Wind Nebula' arXiv:2507.14610